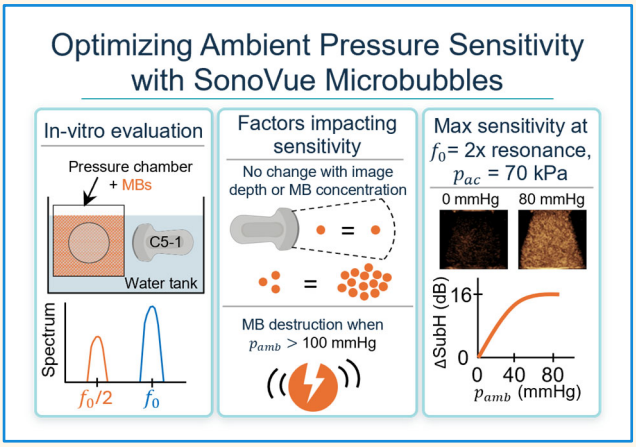
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Implementation of subharmonic imaging on a clinical scanner for optimal ambient pressure sensitivity with SonoVue microbubbles

Lance H. De Koninck, *Graduate Student Member, IEEE*, Hanna Michaelis, Gautam Babu, Jeffry E. Powers, Nico de Jong, and Michalakis A. Averkiou, *Senior Member, IEEE*

***Abstract*— Subharmonic imaging for ambient pressure estimation is a noninvasive alternative to catheter pressure measurements currently used to diagnose multiple disease processes. However, current implementations of subharmonic imaging settings lack important attributes necessary for accurate generation and detection of the subharmonic component. Some implementations employ frequencies near the resonance frequency of microbubbles, requiring high acoustic pressure to generate subharmonic signals. Others utilize short pulses (2-6 cycles) resulting in scattered spectra where the fundamental and subharmonic frequencies overlap. We investigated longer pulses at twice the resonance frequency of SonoVue microbubbles and detected the subharmonic component as a method sensitive to ambient pressure changes. We used an in-vitro pressure chamber filled with SonoVue to vary the ambient pressure (0-140 mmHg) while collecting subharmonic images. We considered multiple frequencies, pulse lengths, acoustic pressures (0-900 kPa), pulsing schemes, imaging arrays, and microbubble concentrations. Native linearized image data and radiofrequency data were used to accurately measure the subharmonic signal. Transmitting at twice the resonance frequency (4.2 MHz) results in the greatest change in subharmonic signal with ambient pressure, a 16 dB increase from 0-80 mmHg (+0.2 dB/mmHg) at 70 kPa acoustic pressure. At low acoustic pressures the sensitivity of subharmonic imaging to ambient pressure changes was not impacted by variation in microbubble concentration or image depth. Higher ambient pressures (> 100 mmHg) resulted in microbubble destruction that reduced image intensity without a change in subharmonic sensitivity. The findings in this work provide evidence of the utility of SonoVue for ambient pressure sensing at low acoustic pressures.**



***Index Terms*— Ambient pressure, clinical scanner, microbubble, ultrasound contrast agent, subharmonic imaging**

I. Introduction

Measurement of physiological pressures is a vital element in the management and diagnosis of multiple disease processes, such as heart failure [11, 12], cirrhosis [13, 14], pulmonary hypertension [15, 16] and traumatic brain injury [17, 18]. Assessment of physiological pressures within the body is generally performed by either direct invasive measurement using a pressure catheter or non-invasive estimation using various modalities such as ultrasound or pressure cuffs. However, catheterization carries procedural risks and is costly [14, 17, 19-22], while non-invasive pressure estimation techniques presently have a narrow scope of assessment and limited degree of accuracy [14, 17, 23-27]. One potential solution that addresses these shortcomings involves the use of ultrasound contrast agents (UCAs) as ambient pressure sensors. When insonified, UCAs produce a nonlinear response that can be separated from the linear signal derived from tissue [28]. Contrast-enhanced ultrasound (CEUS) typically uses a nonlinear pulsing scheme to eliminate the linear, fundamental component of the signal (equal to transmit frequency) and form images with the remaining nonlinear signal (whole harmonics of the transmit frequency) to further isolate microbubble echoes [28]. Another approach is to form contrast images with the subharmonic signal (half the transmit frequency [29]. We note that while tissue can generate

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J. E. Powers is with Philips Ultrasound, Bothell, WA 98021 USA (email: jeff.powers@philips.com).

N. de Jong is with Erasmus Medical Center, Rotterdam, Netherlands (email: n.dejong@erasmusmc.nl).

All other authors are with the Department of Bioengineering, University of Washington, Seattle, WA 98195 USA (e-mail: maverk@uw.edu).

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Highlights

- **Transmitting at twice the resonance frequency results in the greatest change in subharmonic signal with ambient pressure with SonoVue**
- **High ambient pressures resulted in microbubble destruction without a change in subharmonic sensitivity**
- **Optimization of subharmonic imaging at low acoustic pressures eliminates the need for patient-specific acoustic pressures**

harmonics due to nonlinear propagation of ultrasound, only microbubbles generate a subharmonic signal [30, 31] and subharmonic imaging does not suffer from ultrasound propagation artifacts [32, 33]. The subharmonic signal of UCAs has also been shown to be more sensitive to ambient pressure changes than other whole harmonic components [31, 34]. However, disagreement over optimal acoustic parameters, specifically transmit frequency and acoustic pressure, and differences among microbubble contrast agents has led to conflicting information on subharmonic signal change with ambient pressure and delayed clinical translation.

Several in-vitro studies, using scattering experimental set ups and optical measurements, have determined the subharmonic response of microbubbles to be dependent on two acoustic parameters: transmit frequency and incident acoustic pressure. The presence of subharmonic signals has historically been used as an indicator for stable cavitation, sustained oscillations around an equilibrium size, but may not be representative of the complete diversity of cavitation dynamics [35, 36]. However, this type of cavitation requires the use of acoustic pressures (> 200 kPa) that can compromise the microbubble shell and result in the oscillation of a free gas bubble [37, 38]. For lipid-coated microbubbles, the acoustic pressure threshold for subharmonic signal generation is minimized when the driving frequency is twice the average resonance frequency of the microbubble population [39], limiting bubble destruction. Since UCAs, 1-10 μm in diameter [40, 41], are polydisperse with a nonuniform size distribution in either number- or volume-weighted calculations [42, 43], the average resonance frequency (of dilute suspensions) can be calculated by weighted averaging and is typically biased towards the larger end of the distribution. In this regime, microbubbles at low acoustic pressures (5-100 kPa) exhibit a rapid change in effective surface tension of the bubble's shell [44] as it transitions from elastic and buckled states [45], resulting in compression-only behavior [46, 47]. In general, as the acoustic pressure increases (> 200 kPa) with long bursts (> 8 cycles), microbubble shell disruption and lipid shedding will occur, leaving a free gas bubble and shifting the asymmetry of bubble oscillations to favor expansion behavior [44, 45]. These increased pressures also exceed the destruction threshold of certain microbubbles that are resonant at the transmit frequency, indicating that subharmonic signal in this pressure regime is likely due to a combination of collapsing resonant microbubbles and microbubbles oscillating at twice the resonance frequency [37, 47]. As a side note, when using acoustic pressures that result in microbubbles destruction, CEUS becomes more challenging because slow frame rates (typically controlled by triggering, e.g., electrocardiogram triggering) must be used to allow for replenishment of microbubbles [48, 49]. As ambient pressure increases above atmospheric pressure, 3 new effects should take place: a.) inhibition of expansion behavior, b.) increased dissolution rate of free bubbles, and c.) a shift in resonant frequency [50-52]. At low acoustic pressures where we only expect compression behavior with no shell rupture, only the resonance shift need be

TABLE I
SCANNERS, ASSOCIATED TRANSMIT PARAMETERS, AND CONTRAST AGENTS OF IN-VIVO SUBHARMONIC IMAGING STUDIES

Scanner	Probe	Transmit Frequency (MHz)	Transmit pulse length (cycles)	Acoustic pressure range (kPa)[a]	Contrast agents	Reference
Sonix RP	PA4-2	2.5	2[b]	300-900	Definity Sonazoid	[1-3]
SonixTablet	SA4-2[b]	2.5, 3	4, 6[b]	300-600	Definity Sonazoid	[4, 5]
LOGIQ 9	4C	2.5	4[c]	200-900	Definity Sonazoid	[6-10]

[a]Estimated from acoustic pressure range (reported as percentage of maximum acoustic output) used in listed manuscripts after incident acoustic output algorithm. [b]Parameter not specified in all listed manuscripts but assumed to be unchanged based on prior studies by the same research group using the same scanner. [c]Pulse lengths of 2-16 were evaluated in earlier studies, but only the 4-cycle pulse was implemented for later clinical studies.

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considered, while high acoustic pressures must consider all 3 effects. When microbubbles are insonified at low acoustic pressures, an increase in ambient pressure will exaggerate compression behavior due to the low initial surface tension and increase the subharmonic signal [46, 47, 53, 54]. When microbubbles are insonified at high acoustic pressures, increased ambient pressure will dampen the expansion behavior of larger amplitude oscillations and exaggerate dissolution of bubbles with ruptured shells, reducing the number of scatterers. Depending on the balance of these effects, the subharmonic signal will exhibit no change with ambient pressure or even an inverse relationship [55]. Therefore, it is difficult to assign a single value of subharmonic signal change with ambient pressure (dB/mmHg) to a microbubble formulation, leading to the wide range of sensitivities reported in literature.

While subharmonic imaging has been investigated by many groups and has already been commercially released, several questions remain on how to use it, which microbubbles are suited for this mode and why it is not yet clinically used [56]. **Table I** presents the scanners currently known to have (or had) subharmonic imaging capability and used to evaluate ambient pressure in-vivo, their transmit parameters, and the contrast agents used in the listed studies. The scanners include the Sonix RP with a PA4-2 phased array (Ultrasonix Medical Corp., Richmond BC, Canada), SonixTablet (BK Ultrasound, Peabody, MA, USA) with a SA4-2 phased array, and the LOGIQ 9 with a 4C curved array (GE HealthCare, Chicago, IL, USA). All listed studies used transmit frequencies near the average resonance frequency (2.5 MHz) with short pulses. While subharmonic imaging with SonoVue (Bracco, Milan, Italy) has been used in canines for measuring portal hypertension, this was only evaluated at high acoustic pressures (> 300 kPa) using either a single element transducer [57] or a research scanner [58] (transmit frequency 4 MHz, 16 cycles). A more recent release of subharmonic imaging is available on the LOGIQ E10 scanner (GE HealthCare, Chicago, IL, USA) with the C1-6 curvilinear array (implemented with a transmit frequency of 2.5 MHz, receive frequency of 1.25 MHz, 4 cycles, and amplitude modulation) [59], but recent in-vivo studies with this scanner use different probes (C2-9, C1-6-D) where not all transmit parameters are specified, so we excluded this scanner from **Table I** [60, 61]. Although shorter pulses improve the axial resolution of images, these pulses exhibit a wide bandwidth that may result in overlapping fundamental and subharmonic frequencies that cannot be separated. Dave, et al. [7] directly compared 4, 8, and 16 cycle subharmonic imaging pulses in a canine model and found the shortest pulse length (4 cycles) to provide the best correlation of subharmonic amplitudes with portal vein pressures, but spectra of radiofrequency data was not provided so the effect of bandwidth could not be directly observed. Another consideration is by transmitting at 2.5 MHz (and receiving around 1.25 MHz), high acoustic pressures are needed to generate subharmonic signals as this is near the average resonance frequency of UCAs [37]. **Table I** lists the range of acoustic pressures (peak-to-peak) used to evaluate subharmonic amplitude of UCAs in-vivo, estimated from the typical evaluation range of 10-40% of maximum acoustic output as selected by the incident acoustic pressure algorithm [7]. These acoustic pressures generally exceed the acoustic pressure threshold for shell disruption (> 200 kPa) [44, 45]. Consequently, clinical implementation of subharmonic imaging has yet to explore ambient pressure sensing at low acoustic pressures (< 200 kPa) and taking advantage of transmitting at twice the resonant frequency to reduce the subharmonic threshold.

In this study, we implement subharmonic imaging on a clinical scanner and investigate the optimal parameters for noninvasive ambient pressure sensing specifically with SonoVue, a clinically approved microbubble contrast agent. We evaluate the effects of transmit frequency, pulse duration, acoustic amplitude, nonlinear pulsing schemes, and microbubble concentration on the ambient pressure sensitivity of SonoVue. We investigate transmitting acoustic pulses at twice the average resonance frequency to minimize the acoustic pressure threshold for subharmonic signal generation. Furthermore, we evaluate how high static ambient pressures impact subharmonic signal via microbubble destruction. Subharmonic imaging is implemented on a curvilinear and linear array of a clinical scanner across 3 distinct frequency regimes. The acoustic pulses used are longer than conventional fundamental or nonlinear imaging pulses to better segment and detect the subharmonic signals. Three nonlinear pulsing schemes (amplitude modulation-AM, pulse inversion-PI, and amplitude-modulated pulse inversion-AMPI) and single pulses with simple receive filtering are evaluated for producing subharmonic images and maximum ambient pressure sensing. Hydrophone measurements are performed to characterize the acoustic field from different subharmonic imaging modes and measure if sound is directly transmitted at the subharmonic frequency. A custom-built chamber pressurized to physiologically relevant static ambient pressures (0-140 mmHg) is used to evaluate change in subharmonic signal with ambient pressure. Both native linearized image data (with the logarithmic compression removed) and radiofrequency (RF) data are recorded and post-processed to measure changes in image intensity and subharmonic signal magnitude.

II. Material and Methods

A. Implementation of Subharmonic Imaging on Clinical Scanner

Subharmonic imaging was implemented on a Philips EPIQ scanner, C5-1 curvilinear array, and eL18-4 linear array (Philips Healthcare, Bothell, WA, USA). Various combinations of acoustic parameters were programmed (shown in **Table II**), including transmit frequency, receive frequency, number of cycles, nonlinear pulsing schemes and our naming convention. We investigated that transmitting acoustic pulses at twice the average resonance frequency will result in the greatest change in subharmonic signal with ambient pressure compared to other frequencies. Three transmit frequencies were implemented: 2.5 MHz (around resonance), 4.2 MHz (around 2x resonance) and 7.0 MHz (greater than 2x resonance). Due to the available bandwidths in different probes, 2.5 and 4.2 MHz transmit frequencies were implemented on the C5-1 probe, while 7 MHz was implemented on the eL18-4 probe. Receive filters were centered at the corresponding half transmit frequency

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component for the 3 settings (1.25, 2.1, 3.5 MHz,) with a narrow bandwidth to extract only the subharmonic signal. Acoustic pulses were also programmed with 2 pulse lengths multiplied by a Hamming window, 6 cycles representative of conventional CEUS imaging pulse lengths, and windowed 12 cycles to further separate subharmonic and fundamental components. Three standard nonlinear pulsing schemes were implemented to enhance the segmentation of the linear and nonlinear signals : (a) amplitude modulation (AM, transmitting 3 pulses per line, 2 half amplitude and 1 full amplitude, and combining the scattered signal of each pulse at 1/3 amplitude by subtracting the full amplitude from the sum of the half amplitude), (b) pulse inversion (PI, transmitting 2 pulses per line, 1 full amplitude and 1 inverted full amplitude, and adding the scattered signal of each pulse at half amplitude), and (c) amplitude-modulated pulse inversion (AMPI, transmitting 3 pulses per line, 2 half amplitude and 1 inverted full amplitude, and adding the scattered signal of each pulse at 1/3 amplitude). A fourth technique of transmitting a single pulse and using a receive filter around the subharmonic frequency was also evaluated. The produced acoustic pressure is selected by adjusting the output power on the front panel that controls the mechanical index (MI). We note that the $SHI_{AM2.5}$ mode is a standard CEUS setting on the scanner with only the receive filter modified to be centered at the half transmit frequency component with a narrow bandwidth, and is similar to the commercial releases of subharmonic imaging (transmit frequency of 2.5 MHz, receive frequency of 1.25 MHz, 4 cycles, AM pulsing scheme) described in **Table I**. We wanted to have a setting with a small number of cycles to be able to evaluate the effect of pulse length and compare with published results.

TABLE II
MODIFIED TRANSMIT (TX) AND RECEIVE (RCV) SETTINGS FOR SUBHARMONIC IMAGING MODES

Probe	Tx/Rcv Freq. (MHz)	Tx pulse length (cycles)	Pulsing scheme	Mode Name
C5-1	2.5 / 1.25	6	Single AM	$SHI_{2.5}$ $SHI_{AM2.5}$
C5-1	4.2 / 2.1	12	Single AM PI AMPI	$SHI_{4.2}$ $SHI_{AM4.2}$ $SHI_{PI4.2}$ $SHI_{AMPI4.2}$
eL18-4	7.0 / 3.5	12	Single PI	$SHI_{7.0}$ $SHI_{PI7.0}$

The pulse of each subharmonic mode was measured in a water tank with a 0.4 mm membrane hydrophone (UT1604-225; Precision Acoustics, Dorchester U.K.) to evaluate the spectrum of the transmitted sound wave, especially the components at the subharmonic frequency, and map the axial field. **Figure 1** shows hydrophone measurements with $SHI_{2.5}$

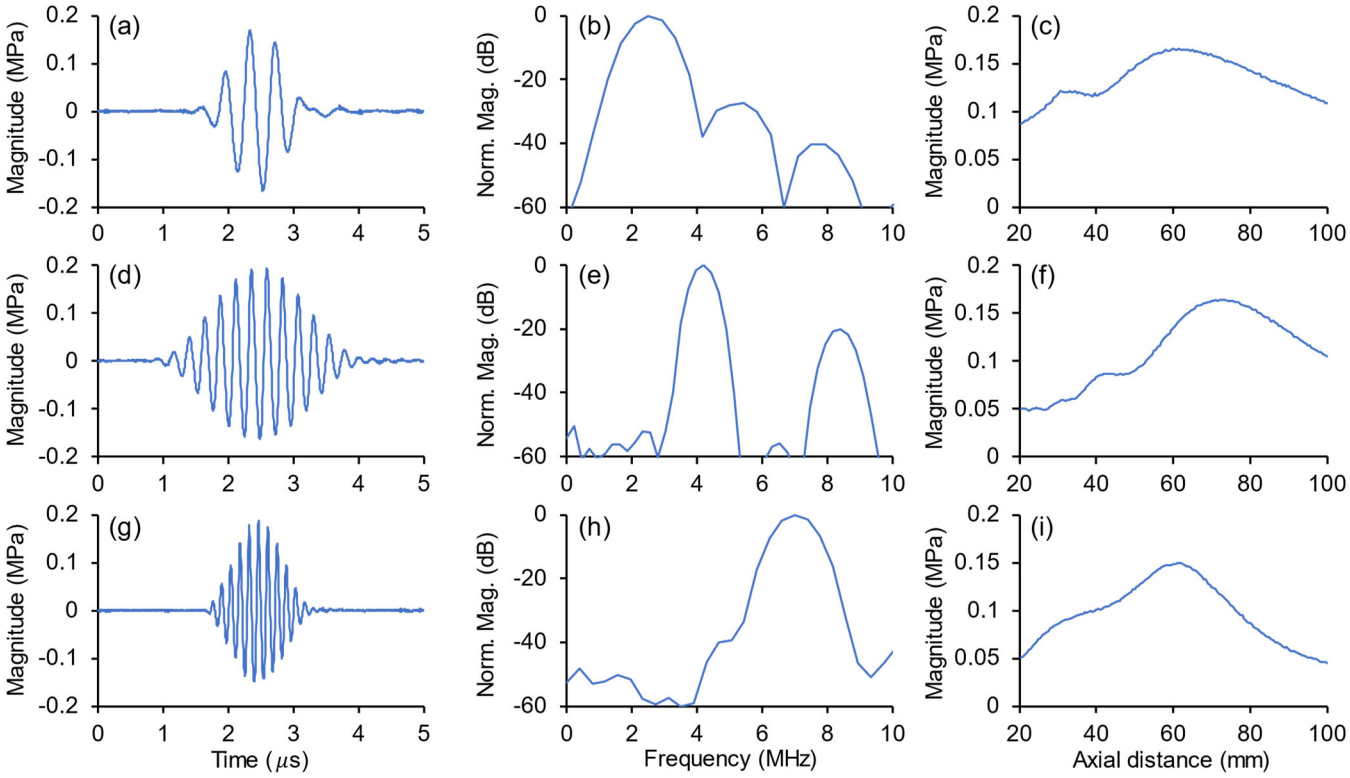


Figure 1. Hydrophone measurements of pulses used in subharmonic imaging modes with the C5-1 probe at 2.5 MHz (top row), 4.2 MHz (center row), and eL18-4 probe at 7 MHz (bottom row). Left column is the transmitted pulse measured at axial distances of (a) 61 mm, (d) 73 mm, and (g) 60 mm. Center column (b, e, h) is the corresponding frequency spectra of time waveforms normalized to the magnitude of the fundamental component. Right column (c, f, i) is the axial pressure variation along the center transmitted line.

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(**Fig. 1(a-c)**), $SHI_{4.2}$ (**Fig. 1(d-f)**), and $SHI_{7.0}$ (**Fig. 1(g-i)**). The left column (**Fig. 1(a), 1(d), 1(g)**) shows the transmitted acoustic pulse measured at the axial location of maximum pressure per subharmonic mode (61 mm for $SHI_{2.5}$, 73 mm for $SHI_{4.2}$, 60 mm for $SHI_{7.0}$). **Figure. 1(b)** shows the overlap of fundamental and subharmonic with $SHI_{2.5}$. The amplitude at the subharmonic component is 20dB below the fundamental and it is not possible to know what part of it belongs to the fundamental and what to the subharmonic. No appreciable acoustic amplitude was measured at the subharmonic component with $SHI_{4.2}$ (**Fig. 1(e)**) and $SHI_{7.0}$ (**Fig. 1(h)**). The right column (**Fig. 1(c), 1(f), 1(j)**) shows the axial pressure distribution measured along the center transmitted line. The azimuthal focus of each subharmonic mode was selected such that the maximum acoustic pressure was delivered 6-7 cm axially from the selected probe. The acoustic pressures reported in this study correspond to the peak negative pressure measured at the axial location of maximum pressure in water during the full amplitude pulse of each pulsing scheme.

B. Experimental Setup

To evaluate the effect of the ambient pressure on the subharmonic signal, we used a custom air-tight chamber capable of pressurizing to 140 mmHg above atmospheric pressure (0 mmHg). A schematic of this setup is shown in **Fig. 2(a)**. The chamber (length = 10 cm, width = 10 cm, height = 10 cm) is submerged in a water tank, with the outer wall positioned 2 cm away from the imaging probe. The ambient pressure was manually controlled in the range of 0-140 mmHg with an air-filled syringe and confirmed with a pressure gauge ($\pm$ 1 mmHg). Circular holes of 5 cm diameter were cut through each wall of the chamber and sealed with a thin polycarbonate sheet (0.13 mm thickness), which was measured to result in negligible transmission loss. A magnetic stir bar placed inside the chamber was used to continuously stir the microbubble solution during experiments. To minimize reflections, the imaging probe was rotated a small angle (5°) relative to the orthogonal position with the chamber wall. **Figure 2(b)** shows an example subharmonic image (set to the maximum dynamic range of 70 dB) without microbubbles, where the dashed lines represent the walls of the chamber demonstrating no reflections inside the chamber.

C. Microbubble Preparation

SonoVue (Bracco Suisse SA, Geneva, Switzerland), a clinically approved contrast agent, was reconstituted according to manufacturer instructions, resulting in a base concentration of ~10^8 microbubbles(MB)/mL [62], and all measurements were completed within 3 hours of preparing the vial. Prior to each experiment, a new microbubble dilution was prepared by extracting a small volume from the vial, mixing in deionized water for 30 s, and then injecting into the chamber. Microbubble dilutions were prepared at 4 concentrations: $2.5 * 10^2$, $1.0 * 10^3$, $4.0 * 10^3$, and $1.0 * 10^4$ MB/mL. The $1.0 * 10^4$ MB/mL concentration was selected to be representative of clinical concentrations [63] when mixed in 5 L blood. To control for variation from different dilution times observed in literature [64], all experiments began 2 minutes from when agent was extracted from the starting vial. Experiments were repeated for a total of 3 trials per microbubble concentration.

D. Experimental Protocol

To measure subharmonic signal change with ambient pressure, two different approaches were utilized. First, the chamber was set to a static ambient pressure between 0-100 mmHg, allowed to mix for 30s, and then the acoustic pressure was gradually increased. At each acoustic pressure, an image loop was recorded consisting of 5 frames captured at a frame rate of 0.33 Hz (once every 3 seconds) to minimize bubble destruction when evaluating the higher acoustic pressures. The frame rate was adjusted with the "timers" trigger option of the scanner which reduces the frame rate to any desired amount. To avoid signal saturation (very bright intensities) at high acoustic pressures (> 400 kPa), the 2D analog gain of the scanner was

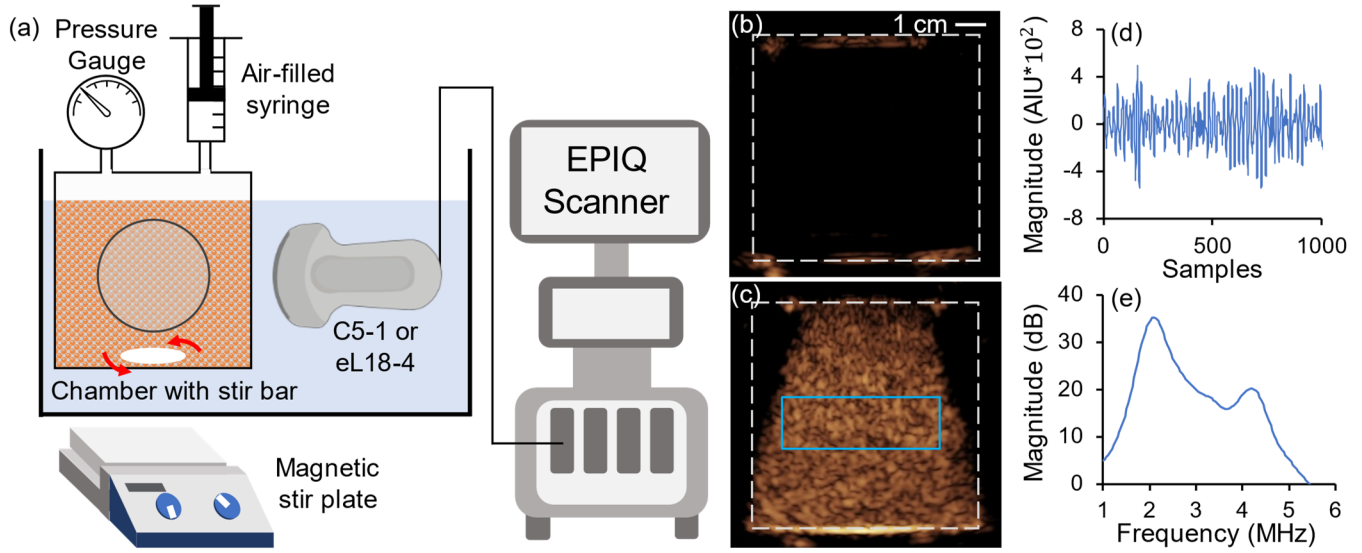


Figure 2. (a) Schematic of the experimental setup. Example subharmonic images (4.2 MHz) of the pressure chamber with (b) no microbubbles, and (c) with SonoVue. The white dashed line outlines the chamber walls and the blue box the region of interest used for image and RF analysis. (d) Example radiofrequency time-series data from a single line after pulse inversion. (e) Example spectra from pulse inversion RF data averaged from 100 transmit lines in dB re 1 arbitrary intensity unit (AIU).

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reduced by 8 dB relative to the gain selected for subharmonic imaging at acoustic pressures < 400 kPa according to the procedure described in Keravnou, et al. [65]. We believe this is a very important point when considering signals with large dynamic range from clinical scanners. During data analysis, image intensity values acquired with high acoustic pressures were adjusted by 8 dB to compensate for the difference in gain. A new microbubble dilution was used for each ambient pressure, with each experiment being completed within 10 minutes of extracting contrast agent from the starting vial. The second approach was to acquire subharmonic images while the ambient pressure varied at a fixed acoustic pressure. This approach allows evaluating any permanent effects of the increased ambient pressure on the microbubble behavior. A single microbubble dilution is imaged during 3 static ambient pressures: 1.) atmospheric pressure (0 mmHg), 2.) increased ambient pressure between 0-140 mmHg (p_{amb}), and 3.) after depressurizing the chamber to return to atmospheric pressure. Image intensities measured when the chamber is not pressurized (p_{amb} = 0 mmHg) are used as a reference to calculate changes in image intensity from ambient pressure-induced microbubble destruction. Each image loop consisted of 5 frames captured at 0.33 Hz frame rate, with loops set 1 minute apart to allow for enough time to record each loop, apply ambient pressure changes, and 30 s of stirring time. RF data was also captured (sampled at 40 MHz) during each p_{amb} image loop, where the RF data from individual pulses that form pulsing schemes is preserved, and saved for post-processing.

E. RF Analysis and Image Quantification

For a subset of experiments, RF data were recorded during p_{amb} image loops and analyzed in MATLAB using a custom script to visualize the frequency spectra of different subharmonic image settings. For the RF data analysis, an image is reconstructed from the raw RF data, and a region of interest (similar to **Fig. 2(c)**) selects data from the center 100 lines. Example of RF data from the center line is shown in **Fig. 2(d)**. The frequency spectrum (dB relative to 1 arbitrary intensity unit-AIU) is then calculated for each line. Finally, all 100 spectra are averaged together to generate a single average spectra per region of interest, an example of which (after PI processing) is shown in **Fig. 2(e)**. To measure subharmonic magnitude from RF data, the average signal within a 0.2 MHz bandwidth centered on the subharmonic component was recorded per spectra.

Native DICOM (linearized image data) were also collected and post-processed with QLAB (Philips Healthcare, Bothell, WA, USA) using a region of interest (**Fig. 2(c)**) spanning 2 cm centered at an image depth of 6 cm to measure linearized image intensity per frame. Two additional regions of interest (ROI) were centered at image depths of 4 cm and 8 cm to evaluate the effect of image depth and acoustic pressure changes on subharmonic sensitivity to ambient pressure. The average image intensity over 5 frames was calculated and coefficient of variation (CV) was used to assess the variation across the 5 image acquisitions per image setting. The change in subharmonic magnitude per unit ambient pressure above atmospheric pressure is used to calculate sensitivity (dB/mmHg). The majority of data analysis presented here is from linearized image data (native Dicom).

III. Results

A. The effect of frequency and bandwidth on subharmonic signal

Figure 3 shows subharmonic images (concentration of $1.0 * 10^4$ MB/mL) at low acoustic pressures (< 200 kPa) during ambient pressures of 0 (atmospheric pressure), 40, and 80 mmHg. In the top row where a short (windowed 6-cycle) pulse at 2.5 MHz in AM mode at 170 kPa is used ($SHI_{AM2.5}$) the image intensity slightly decreases as ambient pressure increases. In the middle row where a longer (windowed 12-cycle amplitude-shaded) pulse at 4.2 MHz in PI mode at 70 kPa is used ($SHI_{PI4.2}$), the image intensity increases with ambient pressure. The same behavior is observed in the bottom row where a similar longer pulse is used at 7 MHz in PI mode at 110 kPa.

The average spectra of RF data of these subharmonic images are shown in **Fig. 4** for the full amplitude pulse (left column) and the combined pulses according to the pulsing scheme (right column). With $SHI_{2.5}$ at 170 kPa, the spectra (**Fig. 4(a)**) exhibit a peak at the transmit frequency (2.5 MHz), and no distinct peak is observed at the subharmonic frequency (1.25 MHz). Additionally, there is no difference between the spectra measured at ambient pressures of 0 and 80 mmHg. The spectra from the AM processing ($SHI_{AM2.5}$) in **Fig. 4(b)** shows a slight reduction (~ 2dB) in magnitude for signal at frequencies between 1 - 4 MHz as ambient pressure increases from 0 to 80 mmHg. Again, no distinct peak is observed at the subharmonic frequency. Conversely, the spectra from $SHI_{4.2}$ at 70 kPa (**Fig. 4(c)**) shows distinct peaks at both the transmit frequency (4.2 MHz) and the subharmonic frequency (2.1 MHz). Increasing ambient pressure from 0 to 80 mmHg results in a 10 dB increase in subharmonic magnitude and no change to the fundamental magnitude. This subharmonic increase with ambient pressure is further increased in the spectra from the PI processing ($SHI_{PI4.2}$, **Fig. 4(d)**), which shows a 14 dB increase in subharmonic magnitude due to the significant reduction in the fundamental magnitude. Similarly, spectra of the $SHI_{7.0}$ pulse at

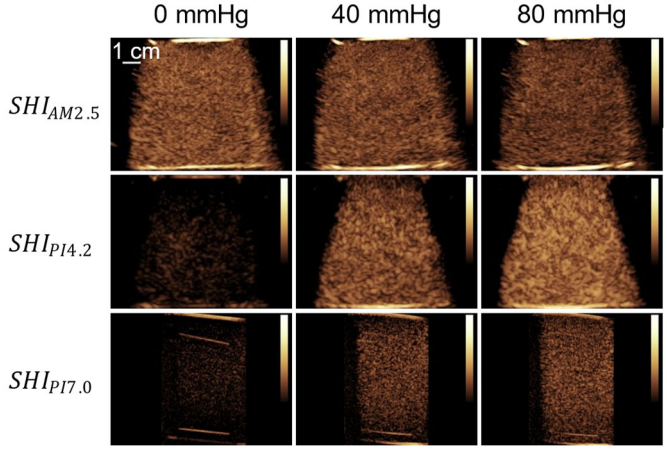


Figure 3. Subharmonic images of SonoVue at low acoustic pressures for 3 ambient pressures. Top row is subharmonic imaging at 170 kPa acoustic pressure with the C5-1 probe ($SHI_{AM2.5}$, 6 cycles at 2.5 MHz transmit and 1.25 MHz receive). Center row is subharmonic imaging at 70 kPa acoustic pressure with the C5-1 probe ($SHI_{PI4.2}$, 12 cycles at 4.2 MHz transmit and 2.1 MHz receive). Bottom row is subharmonic imaging at 110 kPa acoustic pressure with the eL18-4 probe ($SHI_{PI7.0}$, 12 cycles at 7.0 MHz transmit and 3.5 MHz receive).

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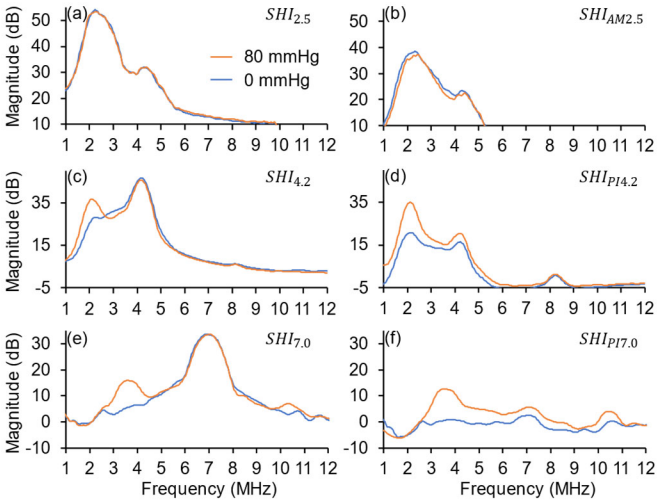


Figure 4. Average spectra of RF data in dB re 1 arbitrary intensity unit, measured during subharmonic imaging of SonoVue. Left column shows spectra of a single pulse and right column shows spectra after applying a pulsing scheme, both at ambient pressures of 0 mmHg (atmospheric) and 80 mmHg. Top row (a-b) uses C5-1 at 2.5 MHz, 6 cycles, amplitude modulation to demonstrate the impact of the transmitted pulse bandwidth. Center row (c-d) uses C5-1 at 4.2 MHz, 12 cycles, pulse inversion with our preferred transmitted bandwidth. Bottom row (e-f) is measured with eL18-4 at 7 MHz, 12 cycles, pulse inversion.

110 kPa (**Fig. 4(e)**) and the combined pulse ($SHI_{PI7.0}$, **Fig. 4(f)**) exhibit peaks at the transmit frequency (7 MHz) and the subharmonic frequency (3.5 MHz) that increases with ambient pressure. Interestingly, we also observe an ultraharmonic component at 10.5 MHz that increases by 4 dB as ambient pressure increases. We note that the RF data analysis presented here was only for demonstrating the frequency spectra of different conditions. The rest of the analysis in this work was done solely on linearized image intensity which is equivalent to the RF observations but easier to perform.

Figure 5 shows subharmonic images of a microbubble solution with a concentration of $1.0 * 10^4$ MB/mL with $SHI_{PI4.2}$ at acoustic pressures of 70, 430, and 700 kPa (rows) and ambient pressures of 0, 40, and 80 mmHg (columns). We note that the images at an acoustic pressure of 70 kPa have an extra gain of 8 dB compared to the images at 430 and 700 kPa. At 70 kPa a

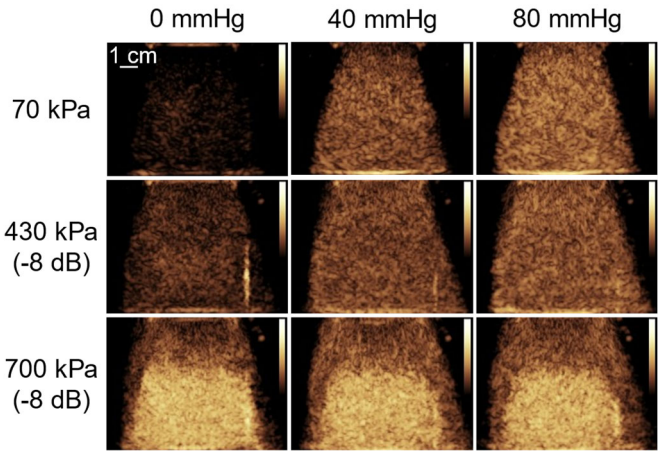


Figure 5. Subharmonic images (acquired with C5-1, 4.2 MHz, 12 cycles, pulse inversion) of SonoVue at 3 ambient pressures and 3 acoustic pressures.

significant increase in subharmonic intensity with ambient pressure is observed. At 430 kPa a smaller increase in subharmonic intensity is observed. Finally at 700 kPa a brighter region appears at an imaging depth of 6-9 cm which does not change with ambient pressure changes. The image brightness at shallower imaging depths (3-5 cm) at 700 kPa still increases slightly with ambient pressure.

Figure 6 shows the average subharmonic intensity of a microbubble solution with a concentration of $1.0 * 10^4$ MB/mL as a function of acoustic pressure for ambient pressures of 0 – 100 mmHg. **Figure 6(a)** shows measurements with $SHI_{AM2.5}$, where the subharmonic intensity generally increases almost linearly with acoustic pressure. At low acoustic pressures (< 200 kPa), the subharmonic intensity is slightly reduced with increasing ambient pressure, and at acoustic pressures greater than 400 kPa subharmonic intensity does not change with ambient pressure. **Figure 6(b)** shows measurements with $SHI_{PI4.2}$, where the subharmonic intensity changes at different rates with acoustic pressure. From 0-160 kPa, the subharmonic intensity rapidly increases, followed by a plateau of subharmonic intensity from 160-350 kPa, and changing again at acoustic pressures greater than 400 kPa where subharmonic intensity increases with acoustic

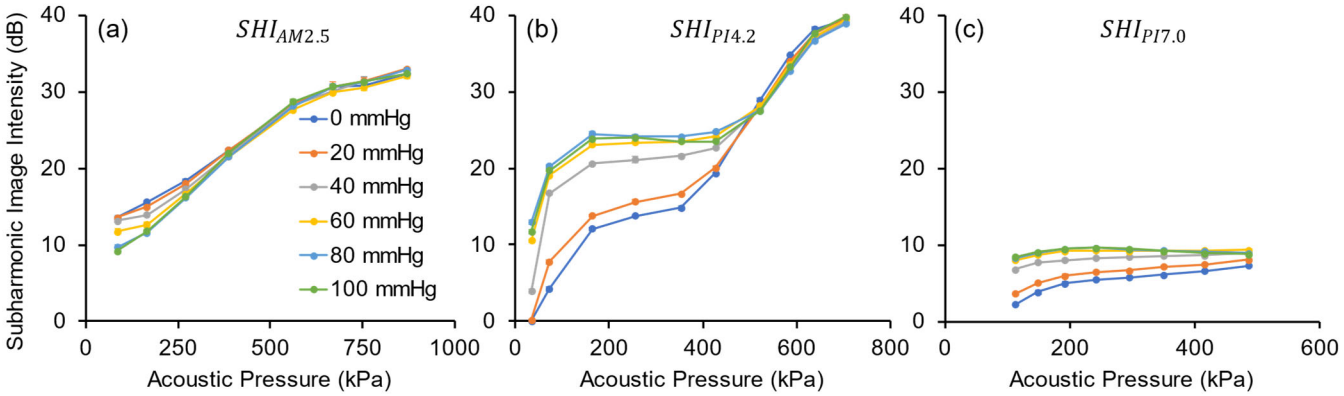


Figure 6. Average image intensity of SonoVue as a function of acoustic pressure for 6 ambient pressures. (a) C5-1 probe, transmit frequency 2.5 MHz, receive frequency 1.25 MHz), 6 cycles, amplitude modulation. (b) C5-1 probe, transmit frequency 4.2 MHz, receive frequency 2.1 MHz, 12 cycles, pulse inversion. (c) eL18-4 probe, transmit frequency 7 MHz, receive frequency 3.5 MHz, 12 cycles, pulse inversion. Error bars represent the standard deviation across 5 image acquisitions.

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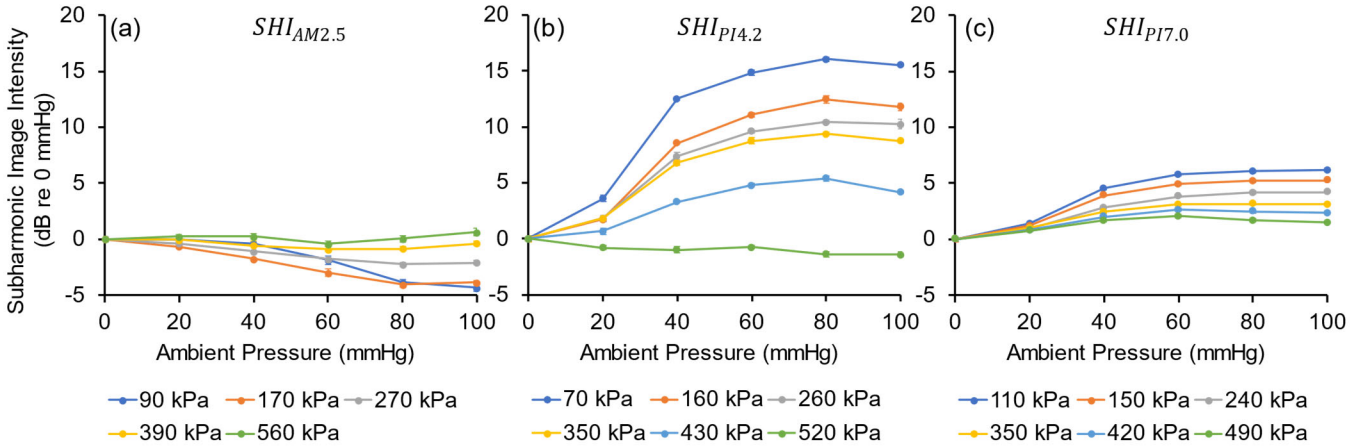


Figure 7. Average image intensity of SonoVue as a function of ambient pressure at different acoustic pressures. (a) C5-1 probe, transmit frequency 2.5 MHz, receive frequency 1.25 MHz), 6 cycles, amplitude modulation. (b) C5-1 probe, transmit frequency 4.2 MHz, receive frequency 2.1 MHz, 12 cycles, pulse inversion. (c) eL18-4 probe, transmit frequency 7 MHz, receive frequency 3.5 MHz, 12 cycles, pulse inversion. Intensity values were normalized with respect to the value at atmospheric pressure (0 mmHg). Error bars represent the standard deviation across 5 image acquisitions.

pressure. When the acoustic pressure is less than 350 kPa, an increase in ambient pressure results in a significant increase in subharmonic intensity, but for acoustic pressures greater than 500 kPa the subharmonic intensity does not change with ambient pressure. At the lowest acoustic pressure of 35 kPa, the measurements at 0 and 20 mmHg did not detect subharmonic signal, while measurements at ambient pressures greater than 40 mmHg detected a subharmonic response. The reason for this is that the intensity measured under 0 and 20 mmHg are below the noise floor. **Figure 6(c)** shows measurements collected with $SHI_{PI7.0}$ where, like $SHI_{PI4.2}$, the subharmonic intensity increases from 0-200 kPa, then plateaus from 200-350 kPa. The subharmonic intensity is also observed to increase with ambient pressure in this acoustic pressure range, with a smaller increase relative to measurements with $SHI_{PI4.2}$.

The same data of **Fig. 6** are plotted as a function of ambient pressure in **Fig. 7** with the different acoustic pressures plotted here as different lines. The subharmonic intensity is normalized to the intensity at 0 mmHg and thus all curves start at 0 dB. For $SHI_{AM2.5}$ (**Fig. 7(a)**), low acoustic pressures (< 200 kPa) exhibit a sensitivity of -0.05 dB/mmHg from 0-80 mmHg. Increasing acoustic pressure reduces this subharmonic signal change, such that no change in subharmonic intensity with ambient pressure is observed at acoustic pressures greater than 400 kPa. For $SHI_{PI4.2}$ (**Fig. 7(b)**), the subharmonic intensity increases with ambient pressure, maximized at an acoustic pressure of 70 kPa with a 16 dB increase and ambient pressure sensitivity of 0.2 dB/mmHg from 0-80 mmHg. As acoustic pressure increases, this sensitivity gradually decreases until no change in subharmonic intensity is observed at 500 kPa. Measurements with SHI_{PI7} (**Fig. 7(c)**) were also maximized at the lowest acoustic pressure (110 kPa) with a 6 dB increase at 80 mmHg and a sensitivity of 0.08 dB/mmHg from 0-80 mmHg. Like $SHI_{AM2.5}$ and $SHI_{PI4.2}$, increasing acoustic pressure decreases the subharmonic sensitivity to ambient pressure. All subharmonic imaging modes exhibited low variation between image acquisitions, with average CV values of 1-2% (maximum 8%).

B. The effect of pulsing schemes on subharmonic signal extraction

Using subharmonic imaging at 4.2 MHz, 4 pulsing schemes (**Table II**) were evaluated at a range of acoustic pressures and ambient pressure with SonoVue (concentration of $1.0 * 10^4$ MB/mL). The subharmonic magnitude detected with each pulsing scheme is highlighted in **Fig. 8**, which shows the average subharmonic intensity as a function of ambient pressure normalized to the subharmonic intensity at 0 mmHg. We note that all pulsing schemes exhibit the same behavior of subharmonic intensity increasing with acoustic pressure as demonstrated with $SHI_{PI4.2}$ in **Fig. 6(b)**. At an acoustic pressure of 70 kPa (**Fig. 8(a)**) all pulsing schemes increase with ambient pressure, maximized at 80 mmHg with a 16 dB increase and sensitivity of 0.2 dB/mmHg from 0-80 mmHg. Small to negligible differences between the pulsing schemes are observed in the 20-60 mmHg ambient pressure range, possibly due to experimental error or noise. PI offers a slightly greater change in subharmonic intensity from 20-40 mmHg, exhibiting a sensitivity of 0.45 dB/mmHg. At the highest acoustic pressure of 700 kPa (**Fig. 8(b)**), all pulsing schemes show no sensitivity to changes in ambient pressure. All pulsing schemes exhibited low

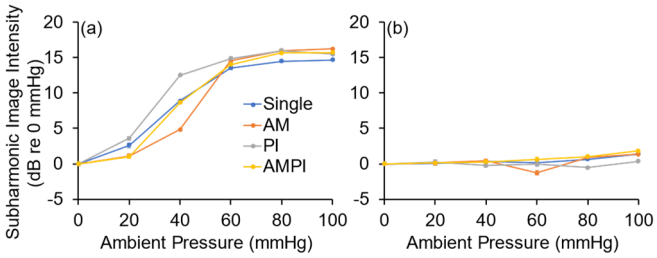


Figure 8. Average image intensity of SonoVue as a function of ambient pressure during subharmonic imaging with the C5-1 probe (transmit frequency 4.2 MHz, receive frequency 2.1 MHz, 12 cycles) with different pulsing schemes at acoustic pressures of (a) 70 kPa and (b) 700 kPa. Intensity values were normalized with respect to the value at atmospheric pressure (0 mmHg). Error bars represent the standard deviation across 5 image acquisitions.

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variation between image acquisitions, with average CV values of 1-2% (maximum 7%).

C. Cycling ambient pressure at 4.2 MHz

To determine if changes in subharmonic intensity were impacted by ambient pressure-induced microbubble destruction, $SHI_{PI4.2}$ images of SonoVue (concentration of $1.0 * 10^4$ MB/mL) were collected at a low acoustic pressure of 120 kPa while cycling ambient pressure between atmospheric pressure and increased ambient pressure, p_{amb}, maintained over a period of time. **Figure 9(a)** shows the average measured subharmonic intensity at each stage of the pressure cycle, as a function of time, for p_{amb} = 0-140 mmHg. When p_{amb} = 0 mmHg (meaning ambient pressure is maintained at 0 mmHg for the whole 2 min duration), the subharmonic signal increases at a rate of 2.2 dB/min (discussed further in Section IV) and an overall increase of 4.4 dB is observed, which we use as the reference subharmonic intensity to determine if microbubble destruction occurred. For 0 mmHg $> p_{amb} \geq$ 80 mmHg, the subharmonic intensity increases by as much as 16 dB during pressurization (t = 1 min) and after depressurizing (t = 2 min) the subharmonic intensity returns to the reference subharmonic intensity (4.4 dB). For $p_{amb} >$ 80 mmHg, the subharmonic intensity increases by a smaller amount at t = 1 min relative to the subharmonic intensity during p_{amb} = 80 mmHg. In addition, for $p_{amb} >$ 100 mmHg, after depressurization (t = 2 min) the subharmonic intensity drops to a value that is lower than before pressurization (t = 0 min). The reduction in subharmonic intensity suggests that SonoVue experiences a permanent change at high ambient pressures (> 100 mmHg), most likely due to microbubble destruction and not due to a change in subharmonic sensitivity to ambient pressure. CV values were consistently low, averaging 2.8% ± 1.4% throughout the entire ambient pressure cycle, indicating no appreciable microbubble destruction occurred during imaging at all ambient pressures.

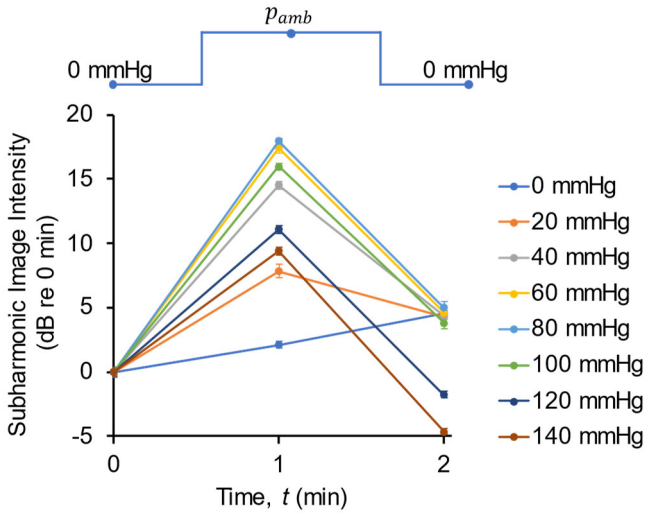


Figure 9. (a) Average image intensity of SonoVue (acquired with C5-1, 4.2 MHz, 12 cycles, pulse inversion, 120 kPa acoustic pressure) as a function of time during prolonged ambient pressure cycling. A schematic of the applied ambient pressure (p_{amb}) as a function of time is shown above the plot. The ambient pressure was 0 mmHg (atmospheric pressure) at t = 0 min, increased to p_{amb} at t = 0.5 min, and decreased back to 0 mmHg at t = 1.5 min. The intensity values were normalized with respect to the value at t = 0 min. Error bars represent the standard deviation across 5 image acquisitions.

D. The effect of microbubble concentration

Figure 10 shows subharmonic images during 70 kPa peak negative acoustic pressure of 3 microbubble concentrations at atmospheric pressure (0 mmHg) and during static ambient pressures of 40 and 80 mmHg. Due to the reduced number of scatterers at lower concentrations, the 2D analog gain of the scanner was increased relative to the reference gain selected for subharmonic imaging with $1.0 * 10^4$ MB/mL shown in **Fig. 3** (decibel values of gain adjustment denoted below each microbubble concentration). While the number of scatterers increases with concentration, all concentrations have similar qualitative increases in subharmonic intensity with ambient pressure.

The average (n = 3) subharmonic intensity with $SHI_{PI4.2}$ as a function of acoustic pressure and ambient pressure at 3 microbubble concentrations is shown in **Figure 11**. We note that the image intensity at 35 kPa acoustic pressure under atmospheric ambient pressure (0 mmHg) was set to a value of 6 dB for each microbubble concentration as a normalization scheme due to differences in 2D analog gain. All microbubble concentrations exhibit the same relationship between subharmonic intensity and acoustic pressure (**Fig. 11(a,b,c)**), in agreement with the trends observed in **Fig. 6b**. All microbubble concentrations also exhibit the same relationship between subharmonic intensity and ambient pressure (**Fig. 11(d,e,f)**), in agreement with the trends observed in **Fig. 7b**. The subharmonic intensity increases with ambient pressure, maximized at an acoustic pressure of 70 kPa for all concentrations. As acoustic pressure increases, subharmonic sensitivity to ambient pressure decreases until no change in subharmonic intensity is observed at 500 kPa for all concentrations.

Figure 12 shows the average (n=3) subharmonic intensity as a function of microbubble concentration for a range of acoustic pressures during static ambient pressures of 0, 40, and 80 mmHg. We note that the image intensities acquired at each

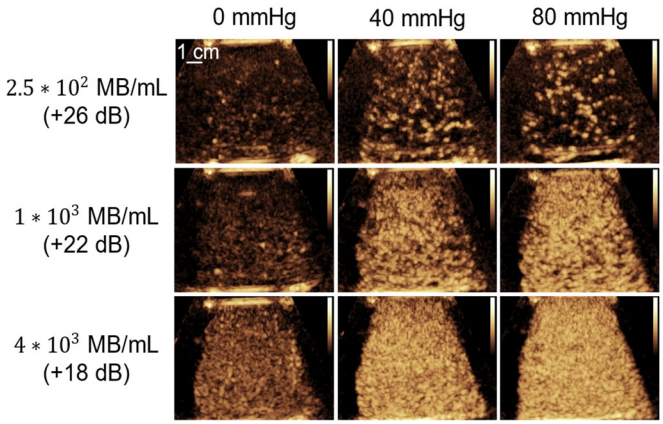


Figure 10. Subharmonic images of SonoVue microbubbles at 3 different concentrations (MB/mL) and 3 ambient pressures. Subharmonic images were recorded at 70 kPa acoustic pressure with the C5-1 probe ($SHI_{PI4.2}$, 12 cycles at 4.2 MHz transmit and 2.1 MHz receive).

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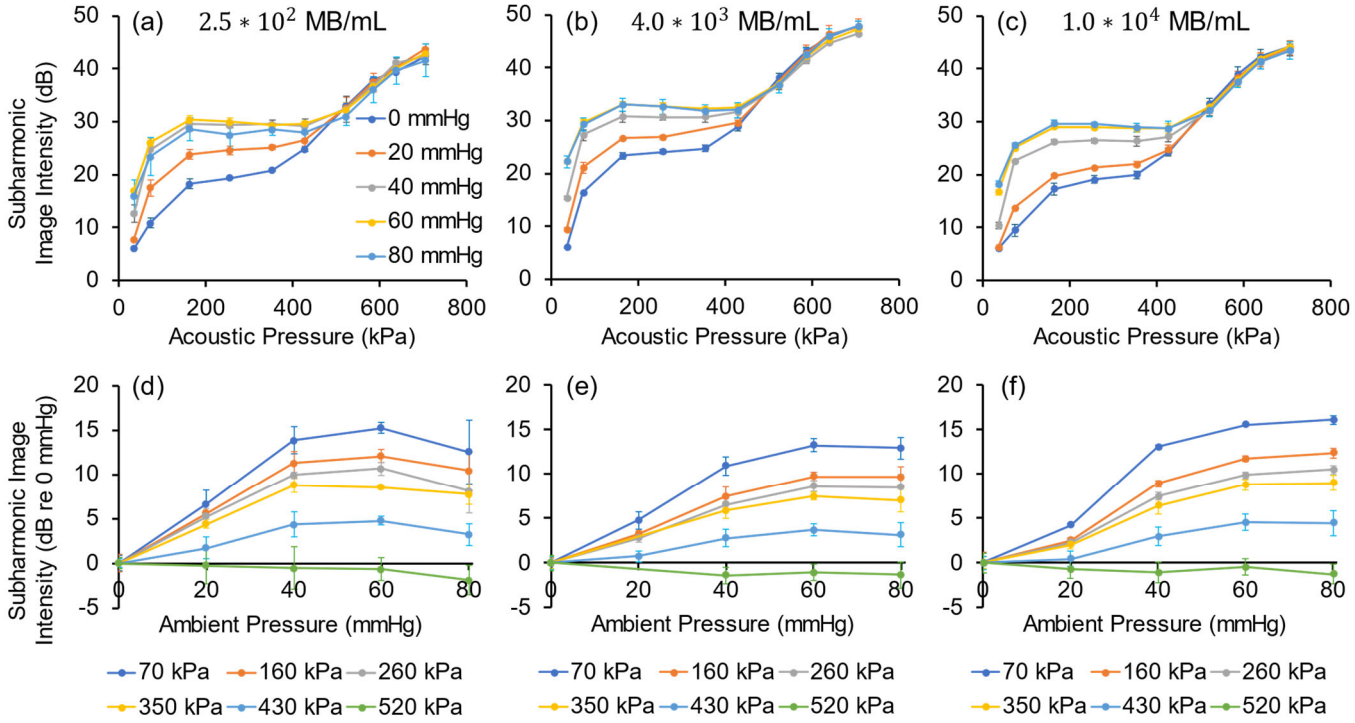


Figure 11. Average (n=3) image intensity of SonoVue as a function of acoustic pressure where different lines correspond to different ambient pressures at 3 microbubble (MB) concentrations: (a) $2.5 * 10^2$ MB/mL, (b) $4.0 * 10^3$ MB/mL, (c) $1.0 * 10^4$ MB/mL. Average image intensity of SonoVue as a function of ambient pressure where different lines correspond to different acoustic pressures at 3 MB concentrations: (d) $2.5 * 10^2$ MB/mL, (e) $4.0 * 10^3$ MB/mL, (f) $1.0 * 10^4$ MB/mL. Subharmonic intensities were acquired with C5-1 probe, transmit frequency 4.2 MHz, receive frequency 2.1 MHz, 12 cycles, pulse inversion. Error bars indicate 1 standard deviation.

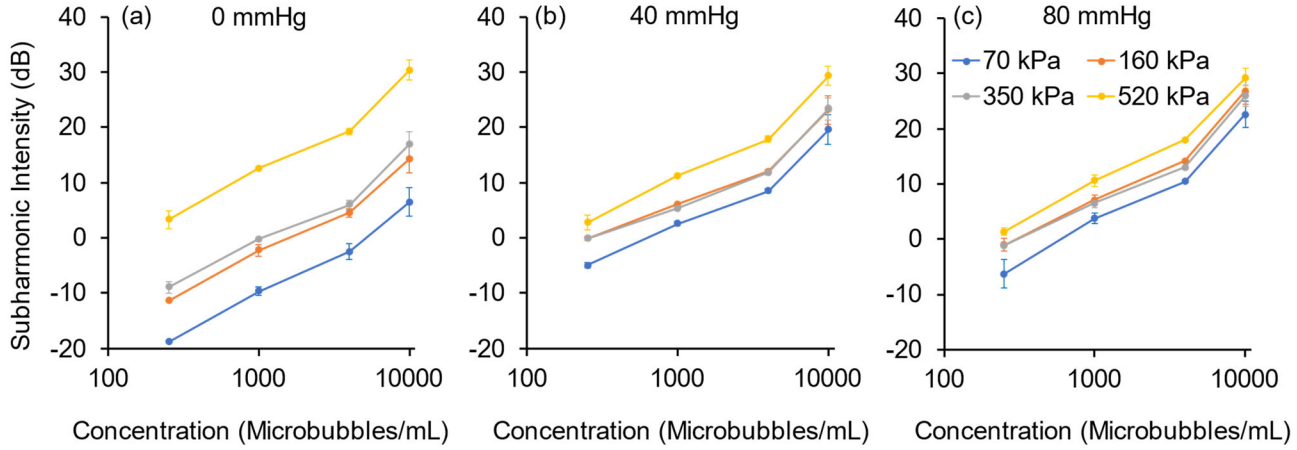


Figure 12. Average (n = 3) image intensity of SonoVue as a function of microbubble concentration at 3 static ambient pressures. Different lines correspond to different acoustic pressures. Subharmonic intensities were acquired with C5-1 probe, transmit frequency 4.2 MHz, receive frequency 2.1 MHz, 12 cycles, pulse inversion. Error bars indicate 1 standard deviation.

microbubble concentration were adjusted by the dB values denoted in **Fig. 10** to compensate for the differences in gain. Subharmonic intensity increased by an average of 16 dB with an order of magnitude increase in microbubble concentration for all ambient and acoustic pressures.

E. The effect of imaging depth

To evaluate the effect of image depth, 3 ROIs centered at 4, 6, and 8 cm were used to measure the subharmonic intensity for a microbubble concentration of $1.0 * 10^4$ MB/mL with $SHI_{PI4.2}$ during 0 mmHg and 80 mmHg. **Figure 13(a)** shows the average subharmonic intensity as a function of image depth for ambient pressure of 0 mmHg and acoustic pressures of 70, 430, and 700 kPa. We note that the incident acoustic pressure in each ROI changes with axial distance and is further discussed in **Section IV-E**. At 70 and 430 kPa, there is little to no change in subharmonic intensity as a function of image depth. At high acoustic pressures (700 kPa), differences in image intensity as a function of depth emerge, where shallow image depths (3-5 cm) exhibit a lower relative intensity compared to deeper image depths (5-7 and 7-9 cm). As ambient pressure increases to 80 mmHg (**Fig. 13(b)**), the trends in image intensity as function of

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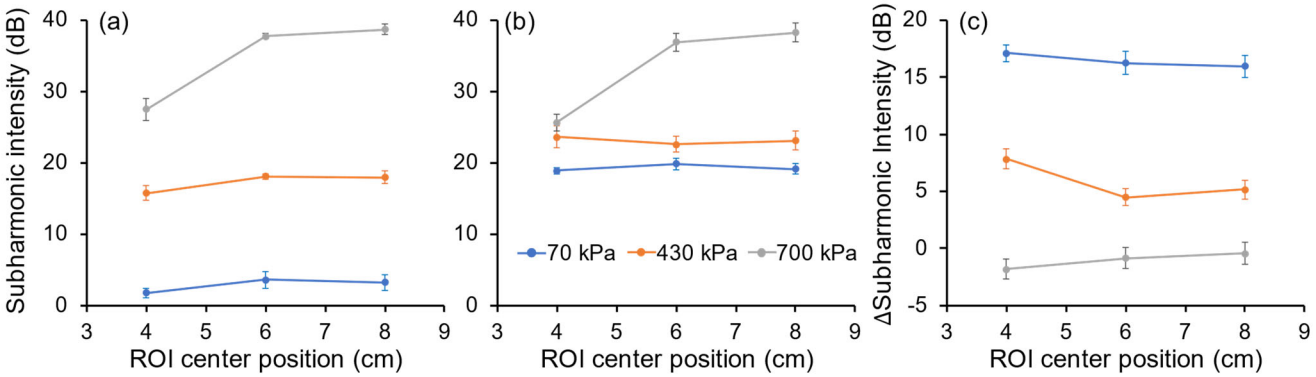


Figure 13. Average (n = 3) image intensity of SonoVue evaluated in different 3 regions of interest (ROI). Different lines correspond to 3 peak acoustic pressures. (a) 0 mmHg, (b) 80 mmHg, and (c) the change in image intensity between 80 mmHg and 0 mmHg. Subharmonic intensities were acquired with C5-1 probe, transmit frequency 4.2 MHz, receive frequency 2.1 MHz, 12 cycles, pulse inversion. Error bars indicate 1 standard deviation.

image depth remain constant, where at low acoustic pressures image depth does not affect image intensity, and shallow image depths exhibit lower intensities at high acoustic pressures. The trends in image intensity as a function of image depth are also in agreement with the qualitative observations shown in the corresponding subharmonic images in **Fig. 5**.

Figure 13(c) shows the relative change in subharmonic intensity between the measurements at 80 mmHg (**Fig. 13(b)**) and 0 mmHg (**Fig 13(a)**). At the lowest acoustic pressure (70 kPa), all ROIs measure an increase of 16-17 dB, in agreement with similar measurements in **Fig. 7(b)** and **Fig. 11(f)**. At 430 kPa, overall sensitivity is reduced compared to 70 kPa, and small differences in subharmonic magnitude change are observed, most likely due to difference in incident acoustic pressure across image depth. Finally at the highest acoustic pressure there is no longer a sensitivity to ambient pressure, with all image depths showing no change in subharmonic magnitude and a value around zero.

IV. Discussion

In this study, we evaluated the effect of transmit frequency, bandwidth, pulsing scheme, and acoustic pressure on the subharmonic response of SonoVue microbubbles under ambient pressures of 0-140 mmHg. We selected to only evaluate SonoVue because of its extensive global adoption and at the same time has not been adequately evaluated for subharmonic-aided pressure estimation (SHAPE). We placed emphasis on low MI subharmonic imaging based on SonoVue's buckling properties that favor low MI subharmonic generation. We covered a comprehensive range of imaging parameters for a total of 8 subharmonic imaging modes (**Table II**) implemented on a clinical scanner and evaluated them in detail in vitro. We evaluated the effect of microbubble concentration on ambient pressure sensing with subharmonics and the effect of image depth (amplitude). We considered the subharmonic response during three transmit frequencies: 2.5 MHz (near resonance), 4.2 MHz (near twice resonance), and 7.0 MHz (above twice resonance). Transmitting at twice the resonance frequency resulted in the greatest change in subharmonic signal with ambient pressure, a 16 dB increase from 0-80 mmHg (+ 0.2 dB/mmHg) at 70 kPa acoustic pressure. High ambient pressures (> 100 mmHg) resulted in microbubble destruction that reduced subharmonic intensity, and not a change in subharmonic sensitivity.

A. Optimal subharmonic imaging mode for ambient pressure sensing for SonoVue

When transmitting near the resonance frequency ($SHI_{AM2.5}$) with small number of cycles (2-4 cycles), little to no change in subharmonic intensity with ambient pressure was observed for all acoustic pressures with SonoVue [see **Fig. 6(a), Fig. 7(a)**]. A small reduction in subharmonic intensity with ambient pressure (-0.05 dB/mmHg) was observed at low acoustic pressures (< 200 kPa), however, average spectra of RF data acquired at 170 kPa acoustic pressure [see **Fig. 4(a), 4(b)**] revealed a considerable overlap between the fundamental and subharmonic signals to the extent that they were not separable. Additionally, hydrophone measurements of the 2.5 MHz full amplitude pulse ($SHI_{2.5}$) showed that due to the wide bandwidth of the fundamental frequency, sound is directly transmitted at the subharmonic frequency [see **Fig. 1(b)**], further confounding the UCA's response to ambient pressure. Our implementation of $SHI_{AM2.5}$ with pulses having only a few cycles was selected such that we could demonstrate the effect of the transmit bandwidth (**Table I**) in order to draw useful comparisons. While studies with the commercially released subharmonic imaging mode (**Table I**) often resulted in good correlation with ambient pressure, in-vitro studies with SonoVue [66], Sonazoid [30, 31, 59, 67, 68], and Definity [31, 59] driven near the resonance frequency have all found ambient pressure sensitivities near our measured sensitivity of -0.05 dB/mmHg with $SHI_{AM2.5}$. Our results with $SHI_{AM2.5}$ simply say that the image intensity in our subharmonic mode decreased with increasing ambient pressure. However, the fact that we (in $SHI_{AM2.5}$) and other similar published works (**Table I**) used short pulses where the subharmonic component is not separable from the fundamental suggests that the observed ambient pressure sensitivity may not be solely due to the subharmonic component. Additionally, in the studies referenced in **Table I**, spectra of RF data are not provided to demonstrate and confirm the extent of subharmonic signal. Subharmonic imaging at the average resonance frequency may still be sensitive to ambient pressure changes, but a narrower transmit bandwidth would improve the detection of the subharmonic signal and enhance

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the sensitivity to ambient pressure. Finally, from a bubble dynamics standpoint, efficient subharmonic generation takes place beyond the initial transient behavior of a bubble [69].

Transmitting at twice the average resonance frequency and using pulses with long cycles ($SHI_{PI4.2}$) resulted in a significant increase in subharmonic intensity with ambient pressure for acoustic pressures less than 400 kPa with SonoVue [see **Fig. 6(b), Fig. 7(b)**]. The subharmonic change with ambient pressure was maximized at an acoustic pressure of 70 kPa, with a sensitivity of 0.20 dB/mmHg from 0-80 mmHg. The relationship between image intensity (subharmonic signal) and acoustic pressure agrees with prior works that have measured the subharmonic response of SonoVue at 3.0 MHz [66] and 4.0 MHz [70]. However, these works relied on single-element transducers with measurements restricted to a small focal area, where we utilized a modified clinical scanner and evaluated a more comprehensive range of acoustic and ambient pressure combinations. While previous studies have reported that the subharmonic generation threshold of lipid shelled microbubbles is dependent on both acoustic and ambient pressure [55, 66, 70], our results indicate that ambient pressures as low as 40 mmHg are sufficient to generate detectable subharmonic signal (dependent on the noise floor of the scanner) even at very low acoustic pressures (35 kPa). Our measured sensitivity is also slightly greater than recent in-vitro evaluations of SonoVue at 4 MHz with low acoustic pressures (120 kPa). Nio, et al. [54] measured a sensitivity of 0.15 dB/mmHg from 0-75 mmHg, and Xu, et al. [70] measured a sensitivity of 0.16 dB/mmHg from 10-40 mmHg. Average spectra of RF data acquired at 70 kPa [see **Fig. 4(c), 4(d)**] showed a distinct subharmonic peak that increases with ambient pressure during both the full amplitude pulse ($SHI_{4.2}$) and after PI processing ($SHI_{PI4.2}$), and the reported image intensities are representative of subharmonic signal changes. We note that the subharmonic component is considered as an "even" harmonic, which PI detects while AM and AMPI detect nonlinear signal at all harmonic components but at a slightly lower level [28]. Small to negligible differences were observed between the nonlinear pulsing schemes at all acoustic pressures [see **Fig. 8**], including the single pulse ($SHI_{4.2}$), suggesting that the narrow bandwidth (12 cycles at 4.2 MHz) used in these subharmonic modes was sufficient to separate the subharmonic and fundamental components. The use of pulsing schemes may still offer some advantages by allowing the use of slightly shorter pulses for improved spatial resolution. However, for our study we conservatively selected longer pulses to ensure good separation of the subharmonic component even without the use of a pulsing scheme. We also note that due to the dependence of ambient pressure sensitivity on acoustic pressure, the subharmonic response with AM and AMPI represents the combined subharmonic behavior across 2 acoustic pressures (half and full amplitude), while PI detects the subharmonic response from a single acoustic pressure. Therefore, the optimal subharmonic imaging mode that maximizes subharmonic signal change with ambient pressure was to transmit at twice the average resonance frequency, required longer than conventional imaging pulses (narrow transmit bandwidth), PI pulsing scheme ($SHI_{PI4.2}$) and a low acoustic pressure of 70 kPa.

Subharmonic imaging above twice the resonance frequency with longer pulses ($SHI_{PI7.0}$) resulted in similar behavior to the $SHI_{PI4.2}$ subharmonic mode, with subharmonic intensity increasing with ambient pressure at acoustic pressures < 400 kPa [see **Fig. 6(c), Fig. 7(c)**], and a distinct subharmonic peak observed in the average spectra of the RF data acquired at lower acoustic pressures (110 kPa) [see **Fig. 4(e), 4(f)**]. However, the detected change in subharmonic signal with ambient pressure (0-80 mmHg) was 0.12 dB/mmHg less with $SHI_{PI7.0}$ (0.08 dB/mmHg at 110 kPa) compared to $SHI_{PI4.2}$ (0.2 dB/mmHg at 70 kPa). One explanation for this reduced ambient pressure sensitivity may be due to the polydispersity (distribution of microbubble sizes) of SonoVue. Despite a similar number-weighted distribution between 0.75 μm radius bubbles (at twice their resonance frequency at 7 MHz [37]) and the average 1.25 μm radius bubbles [40, 42, 43], 0.75 μm bubbles comprise a relatively smaller volume percentage [43], possibly lowering the overall subharmonic signal generated and reducing sensitivity to ambient pressure. We also note that the receive processing of the imaging probe used for $SHI_{7.0}$ (eL18-4) is optimized for imaging depths of 1-4 cm, which may impact signal detection at our selected region of interest from 5-7 cm. While the receive frequency of 3.5 MHz is within the bandwidth of the eL18-4 probe, receiving signal near the edge of the bandwidth may have further reduced the measured sensitivity. Despite the reduced sensitivity, subharmonic imaging at 7 MHz still offers a viable method of detecting ambient pressure changes that can be useful for applications at shallow imaging depths (2-4 cm) such as interstitial fluid pressure measurements of breast cancer [3, 71].

B. Ambient pressure sensitivity of SonoVue

Figure 14 shows the average (n = 3) sensitivity of SonoVue (concentration of $1.0 * 10^4$ MB/mL) as a function of acoustic pressure (calculated from the slopes observed in **Fig. 7(b)**) with $SHI_{PI4.2}$. Sensitivity is shown for 2 ambient pressure ranges, 0-40 mmHg, which is the relevant pressure range for evaluating portal hypertension [72], and 0-80 mmHg, the relevant pressure range for evaluating pulmonary arteries and right heart chambers [73, 74]. SonoVue is sensitive to ambient pressure for acoustic pressures < 500 kPa and is maximized at 70 kPa with 0.33 dB/mmHg from 0-40 mmHg and 0.2 dB/mmHg from 0-80 mmHg. These measurements exceed the maximum sensitivities reported in similar in-vitro studies of SonoVue [54, 59, 66, 70, 75] possibly due to differences in experimental setups and handling of microbubble dilutions [56]. Similar to **Fig. 7(b)**, the sensitivity at 35 kPa is not shown in **Fig. 14** due to no subharmonic signal detected from 0-20 mmHg. Our results also indicate that the optimal acoustic pressure for maximizing ambient pressure sensitivity with SonoVue (oscillating at twice the resonance frequency) is the minimum acoustic pressure (dependent on noise floor of scanner) needed to detect subharmonic signal at atmospheric pressure (70 kPa in this work). A possible mechanism for the maximized sensitivity at low acoustic pressures (< 200 kPa) could be due to SonoVue starting with a tensionless shell where little to no subharmonic signal is generated at atmospheric pressure [44, 47]. In a tensionless state, an increase in ambient pressure induces buckling and compression-only behavior, resulting in

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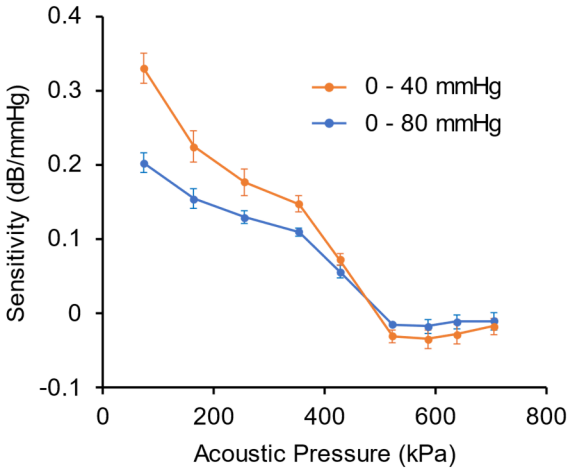


Figure 14. Average (n = 3) ambient pressure sensitivity (acquired with C5-1, 4.2 MHz, 12 cycles, pulse inversion) as a function of acoustic pressure. Ambient pressure sensitivity is calculated from the change in subharmonic signal from 0-40 or 0-80 mmHg using the data presented in Fig. 7(b). Error bars indicate 1 standard deviation.

the generation of subharmonic signal [44, 46, 47]. As acoustic pressure increases, subharmonic signals can be generated by both microbubble oscillation from acoustic excitation and compression from ambient pressure, reducing the sensitivity to subharmonic changes specific to ambient pressure [44, 55]. The increased sensitivity of SonoVue at lower acoustic pressures is an important consideration for clinical implementation of subharmonic imaging (**Table I**) which has relied on an incident acoustic output algorithm to select patient specific acoustic pressures, typically during the "growth" phase of the subharmonic response [7]. However, due to the unique subharmonic response of SonoVue as a function acoustic pressure, confirmed with both our measurements [see **Fig. 6(b)**] and in other in-vitro studies [66, 70], the acoustic output algorithm could be eliminated in favor of a constant, low acoustic pressure (< 200 kPa) that would offer ambient pressure sensitivity with SonoVue and avoid patient-specific calibration. The only necessary adjustment would be small amplitude changes (less than MI~0.1) to account for attenuation and signal-to-noise ratio, similar to conventional CEUS practice.

C. High ambient pressures result in microbubble destruction

After evaluation of subharmonic images ($SHI_{PI4.2}$) during prolonged ambient pressure cycling with SonoVue [see **Fig. 9**], we observed that the high ambient pressures (> 100 mmHg) induce a permanent reduction in subharmonic signal, which we attribute to possible microbubble destruction due to static pressure and not a change in subharmonic sensitivity. When the chamber was not pressurized (p_{amb} = 0 mmHg) and observed for 2 minutes the subharmonic intensity increased by 2.2 dB/min to the reference intensity of 4.4 dB at t = 2 min. This time-dependence of the subharmonic response may be related to the diffusivity of the sulfur hexafluoride gas from SonoVue [40, 76] to the surrounding medium [77] and demonstrates the importance of comparing subharmonic intensities at the same time relative to when microbubbles were injected into the chamber. When the chamber was depressurized after low ambient pressures ($p_{amb} \leq$ 100 mmHg) the subharmonic intensity returned to the reference intensity (4.4 dB) indicating a nondestructive ambient pressure range exists from 0-100 mmHg. During high ambient pressures (p_{amb} > 100 mmHg) the overall subharmonic intensity was reduced, and after depressurizing (t = 2 min) the subharmonic intensity was below the reference intensity, decreased by 6 dB when p_{amb} was 120 mmHg and decreased by 9 dB when p_{amb} was 140 mmHg. A low average CV value was maintained for all data collected in this study, indicating that appreciable microbubble destruction does not occur during imaging. Consequently, any reduction in subharmonic intensity would be due to the high ambient pressure. Our results also agree with Nio, et al. [54] who reported similar descending subharmonic intensity at high ambient pressures (> 125 mmHg) with SonoVue and speculated that the decrease is due to microbubble buckling and destruction. This is important to consider when using subharmonic imaging in patients with severe heart disease, where ambient pressures can exceed 100 mmHg [3] and may require other techniques such as ultrafast imaging to measure dissolution time [52, 78]. The increase in subharmonic signal over time must also be considered for in-vivo work with SonoVue, as clinical implementation of subharmonic imaging (**Table I**) currently utilizes continuous infusions for microbubble delivery which if paired with SonoVue may result in an ambient pressure sensitivity that changes over time. Therefore, subharmonic imaging techniques compatible with bolus injections should be developed to eliminate variation in ambient pressure sensitivity dependent on evaluation time.

D. Effect of microbubble concentration on ambient pressure sensitivity

Our investigation of subharmonic intensity of SonoVue at different microbubble concentrations provides several key findings that may better guide future evaluations of subharmonic sensitivity to ambient pressure. The change in subharmonic intensity with acoustic pressure and ambient pressure [see **Fig. 11**] was similar at each microbubble concentration, where change in subharmonic intensity with ambient pressure was maximized at an acoustic pressure of 70 kPa increasing by 13-16 dB with sensitivities of 0.16-0.2 dB/mmHg from 0-80 mmHg. The similar subharmonic response across microbubble concentrations indicates that differences in delivered contrast agent that may occur in the clinic will not significantly affect subharmonic sensitivity so long as the volume injected remains constant. The small differences in sensitivity across microbubble concentrations are likely due to the necessity of a serial dilution to create the lower concentrations ($2.5 * 10^2$ and $1.0 * 10^3$ MB/mL), introducing greater compounding error and reduced precision compared to the higher concentrations which were prepared via a single-step dilution. The similar subharmonic sensitivity to ambient pressure across microbubble concentrations agrees with Gupta, et al. [30] where the SHAPE gradient measured with Sonazoid in-vivo was found to be independent of microbubble concentration (evaluated from approximately $2.5 * 10^5$-$1.4 * 10^6$ MB/mL). An interesting finding not previously reported is that the subharmonic image intensity scales with microbubble concentration in a very similar fashion to how conventional nonlinear image intensity scales with microbubble

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concentration [79]. Furthermore, this intensity-concentration relationship is maintained at the higher ambient pressures [see **Fig. 12**].

E. Effect of imaging depth on measured subharmonic response of SonoVue

Imaging depth can strongly affect incident acoustic pressure due to diffraction/focusing and signal-to-noise ratio which can influence the measured subharmonic response [28]. To date, no subharmonic imaging studies have directly measured the effects of imaging depth, and how the placement of a region of interest can influence the measured subharmonic sensitivity to ambient pressure. In this work we demonstrate for SonoVue imaged with the optimal subharmonic imaging mode ($SHI_{PI4.2}$) at acoustic pressures of 70 and 430 kPa the subharmonic signal does not change with imaging depth, and at higher acoustic pressure (700 kPa) the ROI depth affects the measured subharmonic intensity [see **Fig. 13**]. Using the axial pressure distribution for $SHI_{PI4.2}$ [see **Fig. 1(f)**] we can also estimate the incident acoustic pressure within each ROI, calculated as the maximum acoustic pressure within the boundaries of the ROI. When the ROI is at the shallowest image depth of 3-5 cm, the maximum acoustic pressure is approximately 55% of the peak negative pressure (measured at an axial distance of 7 cm), corresponding to actual acoustic pressures of 40, 240, and 400 kPa instead of 70, 430 and 700 noted in **Fig. 13**. The deeper ROIs positioned at 5-7 cm and 7-9 cm both overlap with the maximum axial acoustic pressure at 7 cm, resulting in similar acoustic pressures between the 2 regions of 70, 430, and 700 kPa in **Fig. 13**. The consistent subharmonic signal across imaging depths at low acoustic pressures may be due to acoustic pressure response of SonoVue [see **Fig. 11(a-c)**], where subharmonic signal remains constant from 200 to 400 kPa which may offset depth-dependent variations in incident acoustic pressure. Additionally, the high microbubble concentration ($1.0 * 10^4$ MB/mL) did not produce significant depth-dependent attenuation, with measured subharmonic signal remaining constant across image depth at low peak pressures ($\leq$ 430 kPa) and increasing with image depth at 700 kPa. This implies no acoustic shadowing was present at this concentration [28]. The depth-independent response observed at low acoustic pressures (< 430 kPa) further supports low acoustic pressure insonation as an optimal strategy for maximizing the subharmonic sensitivity of SonoVue to ambient pressure, with potential improvement from transmit beamforming that produces a more uniform acoustic pressure distribution.

F. Limitations of in-vitro evaluation

In this study we used a pressurized chamber placed in a water tank to evaluate the subharmonic response SonoVue and several subharmonic imaging modes. This allowed for direct assessment of freshly prepared microbubbles covering a wide range of acoustic conditions to accurately quantify the change in subharmonic signal to ambient pressure. However, the simplicity of in vitro setups eliminates some factors that are present in-vivo. Microbubbles were injected directly from the prepared dilution into the chamber without first filtering the dilution to remove any large microbubbles, a function performed by the lungs in-vivo [80]. While the number of large bubbles only constitute a small percentage of the microbubble population [40, 41], the presence of these microbubbles could potentially influence the results in-vitro. Another factor was the absence of flow in our experimental setup, requiring a slow frame rate (0.33 Hz) to protect microbubbles from destruction during repeated scanning at the higher acoustic pressure. Still, our measurements of the subharmonic response of SonoVue as a function of acoustic pressure [see **Fig. 6(b)**] indicate that sensitivity to ambient pressure is maximized at nondestructive acoustic pressures below the threshold for microbubble shell rupture (< 200 kPa) [see **Fig. 14**] [44, 45]. By optimizing subharmonic imaging at low acoustic pressures, clinical implementation of $SHI_{PI4.2}$ can be performed with real-time frame rates and does not require triggering methods (e.g., electrocardiogram triggering) that are necessary for high acoustic pressure techniques [48, 49]. Additionally, the presence of tissue in-vivo introduces the effect of attenuation which is dependent on both imaging depth and transmit frequency. We note that the effect of attenuation may impact detected subharmonic signal during in-vivo evaluations. However, our ROI imaging depth analysis demonstrates that the impact of amplitude variations will be minimal at low acoustic pressures.

V. Conclusion

In this study, we evaluated the effect of 3 different transmit frequencies (near resonance, twice resonance and greater than twice resonance), bandwidth, pulsing scheme, and acoustic pressure on the subharmonic response of SonoVue microbubbles under ambient pressures of 0-140 mmHg. Transmitting at twice the resonance frequency (4.2 MHz) results in the greatest change in subharmonic signal with ambient pressure, a 16 dB increase from 0-80 mmHg (+ 0.2 dB/mmHg) at 70 kPa acoustic pressure, greater than the maximum sensitivities reported in similar in-vitro evaluations of SonoVue. By implementing subharmonic imaging at low acoustic pressures (< 200 kPa) similar to conventional CEUS practice, SonoVue driven at twice its resonance frequency will be at its maximum sensitivity to ambient pressure, eliminating the need for acoustic pressure optimization used in some reported approaches over a wide range of acoustic pressures. At acoustic pressures less than 430 kPa, the sensitivity of SonoVue to ambient pressure changes is not impacted by variation in microbubble concentration or imaging depth when using $SHI_{PI4.2}$. Prolonged exposure to high ambient pressures (> 100 mmHg) resulted in microbubble destruction that impacts subharmonic intensity. The findings in this work provide evidence of the utility of SonoVue for ambient pressure sensing at low acoustic pressures.

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Lance H. De Koninck (Member, IEEE) was born in Sandy, UT, USA, in 1998. He received the B.S. degree in engineering science and mechanics from Virginia Tech, Blacksburg, VA, USA in 2020. He is currently pursuing the Ph.D. degree in bioengineering from University of Washington, Seattle, WA, USA under the mentorship of Dr. Michalakis A. Averkiou.

His research interests include therapeutic applications of ultrasound and microbubbles such as enhanced drug delivery with ultrasound-mediated microbubble cavitation and ultrasound-mediated mild hyperthermia.



Jeffrey E. Powers was born in Santa Monica, California, in 1950. He received a BS in Physics and MS in Bioengineering from UC San Diego in 1972 and 1974 respectively. He completed a PhD in Electrical Engineering at the University of Washington in 1980 specializing in ultrasonic array signal processing.

He spent a year in Basel, Switzerland. on a post doc fellowship building a device to study platelet shape changes using laser light scattering. He has worked for ATL/Philips Ultrasound since 1982 and was the principal developer of their color flow feature. Since 1992 he has been largely involved with ultrasound contrast and nonlinear acoustics, managing the development of the first harmonic imaging system for tissue and contrast, and leading the development of other new contrast imaging features. Most recently he has been leading the development of ultrasound system modifications to support research in the use of contrast microbubbles for minimally invasive therapy, including stroke, myocardial infarction, drug and gene delivery, and microvascular flow enhancement. Throughout his career he has collaborated with many technical and clinical academic centers on a variety of research projects. He has coauthored many publications and more than 30 patents.



Michalakis A. Averkiou (M'99-SM'11) was born in Nicosia, Cyprus, in 1963. He received the B.S., M.S., and Ph.D. degrees in mechanical engineering with specialization in nonlinear acoustics/biomedical ultrasound from the University of Texas at Austin, Austin, TX, USA, in 1987, 1989, and 1994, respectively.

He was a Postdoctoral Fellow with the Applied Physics Laboratory, University of Washington, Seattle, WA, USA, from 1994 to 1996. He was with Philips Medical Systems, Bothell, WA, USA, from 1996 to 2005, and worked on diagnostic ultrasound imaging, specifically tissue harmonic imaging, and ultrasound contrast agents. He was an Associate Professor with the Department of Mechanical and Manufacturing Engineering at the University of Cyprus from 2005 to 2015. He then joined the Department of Bioengineering at the University of Washington, first as an Associate Professor from 2015-2022, and then as a Professor since 2023.

Dr. Averkiou is currently an Acoustical Society of America Fellow and was awarded the Marie Curie Chair of Excellence from the European Commission.

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